

WORK PROCEDURE

PRO-OP-ITH-016A

PROCEDURE FOR HANDLING ITH HAMMER AND COMPONENTS

OBJECTIVE

The purpose of this procedure is to define the sequence of operations for handling the drill hammers and components. This includes handling, installing the hammer, adding or removing bits and adding the "DRIVER SUB".

APPLICATION

This procedure applies to all employees of MACHINES ROGER INTERNATIONAL.

INTERPRETATION

The person having the authority for the interpretation of this document is the Operations Manager. Any request for modification, revision or update must be made to this person.

RESPONSABILITIES

The driller is responsible for ensuring good communication with the driller-helper. They must follow this procedure. The supervisor must enforce the procedure.

CONTEXT

The risks associated with handling long-hole hammers are related to Musculoskeletal Disorders (MSD) as well as injuries from falling hammers. Due to the weight of these hammers, bad working methods can cause accidents with serious consequences. To control these risks, a transport tool exists, and work sequences must be followed to avoid exposing workers to a fall from a hammer, or the development of MSDs.

The weight of the hammers according to supplier specifications:

Hammer 3"	60 lbs
Hammer 4"	89 lbs
Hammer 6 "	211 lbs

Sources : Sandvik and Epiroc

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THE 3 "AND 4" HAMMERS MUST BE INSTALLED ON THE GEAR BOX MANUALLY IN THE SAME MANNER AS THE DRILL RODS. THE WORKER WILL THEN APPLY THIS PROCEDURE FROM POINT # 8.

6 "HAMMERS MUST ALWAYS BE TRANSPORTED USING THE HAND TRUCK DOLLY DESIGNED FOR THIS PURPOSE. THE "DRIVER SUB" MUST BE INSTALLED ON HAMMERS AT ALL TIMES DURING HANDLING



To screw the 6 " hammer into the gear box:

- 1- Approach the hand truck with the hammer near the drill;
- 2- Open the tables;
- 3- Position the hammer between the tables under the gear box using the hand truck;
- 4- Lower the GEAR BOX and align the "SAVER SUB" against the threads of the hammer;
- 5- Turn the head to screw the hammer into the "SAVER SUB" until the threads are butted;
- 6- Remove the safety chain on the hand truck;
- 7- Reassemble the head with the hammer and remove the hand truck;

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8- Lower the hammer back into the jaws and close them on the hammer, then close the tables;

9- Tighten the threads making sure that they are well butted;

10-Apply the spline piston;



To install a 6 ", 8" or 10 "bit on the 6" hammer;

EVERY TIME YOU HAVE TO EXPOSE YOUR HANDS UNDER THE HAMMER, IT MUST BE VALIDATED THAT THE HAMMER IS TIGHTENED INTO THE GEAR BOX

11-Install the bit-breaker in the slip plate and close the jaws;



12-Place the bit on the bit breaker;



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13-Lower the hammer on the bit and unscrew the driver sub with the rotation



14-Then mount the hammer high enough to allow the "RETAINING RING" to be installed;

15-Lower the hammer to the driver sub and screw them together with the rotation so that they are properly threaded;

16-Raise the hammer enough to remove the bit breaker;

17-Start drilling;

To remove a 6", 8" or 10" bit on the 6" hammer

EVERY TIME YOU HAVE TO EXPOSE YOUR HANDS UNDER THE HAMMER, IT MUST BE VALIDATED THAT THE HAMMER IS TIGHTENED INTO THE GEAR BOX

18-Raise the hammer high enough to be able to install the bit breaker in the slip-plate and close the jaws;



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19-Lower the hammer until the bit is in the bit breaker;



20-Break the hammer and unscrew from only one turn;

21-Make sure the spline-piston is applied ;



22-Break the bit and unscrew the hammer from the driver sub ;

23-Raise the hammer high enough to remove the retaining rings;

24-Remove the retaining rings;

25-Lower the hammer on the driver sub and screw them together;



26-Raise the hammer high enough to remove the bit and the bit breaker;

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27-Open the jaws;

28-Release the spline piston



29-Lower the hammer in the jaws and close them on the hammer;

30-Open the jaws;

31-Raise the hammer to put the hand truck under the tables ;

32-Lower the hammer over the hand truck, leaving space to finish unscrewing the hammer;

33-Unscrew the hammer manually without exposing hands;

34-Attach the hammer using the retaining chain and remove the hand truck.



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